



Linux Commands Documentation

A beginner-friendly cheat sheet for Linux, Docker, Kubernetes, AWS & DevOps

1. Navigation

Command	Meaning	Example
<code>pwd</code>	Show current directory	<code>pwd</code>
<code>ls</code>	List files/folders	<code>ls</code>
<code>ls -l</code>	Detailed list	<code>ls -l</code>
<code>ls -a</code>	Show hidden files	<code>ls -a</code>
<code>cd</code>	Change directory	<code>cd Documents</code>
<code>cd ..</code>	Go one directory back	<code>cd ..</code>
<code>cd ~</code>	Go to home directory	<code>cd ~</code>
<code>cd /</code>	Go to root directory	<code>cd /</code>

Example

```
pwd
ls
cd Documents
ls
cd ..
```

2. Files and Directories

Command	Meaning
<code>mkdir</code>	Create directory
<code>touch</code>	Create empty file
<code>cp</code>	Copy
<code>mv</code>	Move or rename
<code>rm</code>	Delete
<code>rmdir</code>	Remove empty directory

Examples

Create folder:

```
mkdir project
```

Create file:

```
touch app.py
```

Copy:

```
cp app.py backup.py
```

Move:

```
mv app.py project/
```

Rename:

```
mv old.txt new.txt
```

Delete:

```
rm app.py
```

Delete directory:

```
rm -r project
```

⚠ Be careful with `rm -rf .`

3. Reading Files

Command	Meaning
<code>cat</code>	Display file
<code>less</code>	Read large file
<code>head</code>	Show beginning
<code>tail</code>	Show end
<code>tail -f</code>	Continuously watch changes

```
cat file.txt
head file.txt
tail file.txt
```

For logs:

```
tail -f app.log
```

4. Searching

grep

Search for text:

```
grep "error" app.log
```

Search recursively:

```
grep -r "password" .
```

find

Find files:

```
find . -name "app.py"
```

Find directories:

```
find . -type d -name "backend"
```

5. Permissions

Linux has permissions for:

```
Owner
Group
Others
```

Check permissions:

```
ls -l
```

Example:

```
-rwxr-xr-x
```

chmod

Change permissions:

```
chmod +x script.sh
```

Now you can run:

```
./script.sh
```

chown

Change file owner:

```
sudo chown user:user file.txt
```

6. **sudo**

sudo runs a command with administrator/root privileges.

```
sudo apt update
sudo systemctl restart nginx
```

Think:

```
sudo = "Run this as administrator"
```

7. Package Management — Ubuntu

Update package information:

```
sudo apt update
```

Upgrade packages:

```
sudo apt upgrade
```

Install:

```
sudo apt install docker.io
```

Remove:

```
sudo apt remove docker.io
```

Search:

```
apt search docker
```

8. Processes

ps

Show processes:

```
ps
```

All processes:

```
ps aux
```

```
top
```

Live process monitor:

```
top
```

```
kill
```

Stop a process:

```
kill 1234
```

Force stop:

```
kill -9 1234
```

Here `1234` is the **PID (Process ID)**.

Find a process:

```
ps aux | grep node
```

9. Disk and Memory

Disk space:

```
df -h
```

How much disk space is available?

Folder size:

```
du -sh project/
```

How much space does this folder use?

RAM:

```
free -h
```

CPU information:

```
lscpu
```

10. Networking

These are particularly important for DevOps.

IP address:

```
ip a
```

Show my network interfaces and IP addresses.

Network interfaces:

```
ip link
```

Routing:

```
ip route
```

Show where network traffic goes.

Test connectivity:

```
ping google.com
```

Show listening ports:

```
ss -tuln
```

Test a website/API:

```
curl https://example.com
```

Download:

```
wget https://example.com/file.zip
```

Show local IP:

```
hostname -I
```

DNS lookup:

```
nslookup google.com
```

Detailed DNS lookup:

```
dig google.com
```

Network management:

```
nmcli device status
```

11. Telnet

`telnet` can be used to test whether a particular port is reachable.

```
telnet localhost 5000
```

Can I connect to port 5000?

For modern systems, `nc` is often more convenient:

```
nc -zv localhost 5000
```

12. Services

Ubuntu uses `systemctl` to manage services.

Check:

```
sudo systemctl status nginx
```

Start:

```
sudo systemctl start nginx
```

Stop:

```
sudo systemctl stop nginx
```

Restart:

```
sudo systemctl restart nginx
```

Enable at startup:

```
sudo systemctl enable nginx
```

Disable:

```
sudo systemctl disable nginx
```

13. Environment Variables

Show a variable:

```
echo $PATH
```

Create a temporary variable:

```
export PORT=5000
```

Check:

```
echo $PORT
```

Show all environment variables:

```
env
```

14. Archives

Create tar.gz:

```
tar -czvf backup.tar.gz project/
```

Extract:

```
tar -xzvf backup.tar.gz
```

ZIP — Create:

```
zip -r project.zip project/
```

ZIP — Extract:

```
unzip project.zip
```

15. Users

Current user:

```
whoami
```

Logged-in users:

```
who
```

User information:

```
id
```

Change user:

```
su username
```

16. System Information

Operating system:

```
uname -a
```

Linux distribution:

```
cat /etc/os-release
```

Hostname:

```
hostname
```

Current date:

```
date
```

17. History

Show previous commands:

```
history
```

Run a previous command:

```
!100
```

where 100 is the history number.

Clear terminal:

```
clear
```

18. Redirection

> — Write output to a file:

```
echo "Hello" > file.txt
```

>> — Append to a file:

```
echo "World" >> file.txt
```

| — Send output of one command to another:

```
ps aux | grep nginx
```

Think: Command 1 → Command 2

19. Useful Shortcuts

Shortcut	Meaning
Ctrl + C	Stop running command
Ctrl + L	Clear terminal
Ctrl + D	Exit shell/session
Ctrl + Z	Pause process
Tab	Auto-complete
↑	Previous command
↓	Next command
Ctrl + R	Search command history

20. Linux Command Help

You don't need to memorize every command.

`man`

Linux documentation:

```
man ls
man chmod
```

Exit with: `q`

`--help`

Quick help:

```
ls --help
docker --help
```

☆ Most Important Commands for You

Since you're learning **Docker + Kubernetes + AWS/DevOps**, prioritize these:

Navigation: `pwd`, `ls`, `cd`

Files: `mkdir`, `touch`, `cp`, `mv`, `rm`

Reading/Search: `cat`, `less`, `head`, `tail`, `grep`, `find`

Permissions: `chmod`, `chown`, `sudo`

Processes: `ps`, `top`, `kill`

Networking: `ip a`, `ip route`, `ping`, `ss`, `curl`, `wget`, `telnet`, `nc`, `nslookup`, `nmcli`

System: `df`, `du`, `free`, `lscpu`

Services: `systemctl`

Packages: `apt`

Archives: `tar`, `zip`, `unzip`

One important rule

When you forget a command, don't immediately search Google. Linux itself provides documentation:

```
man <command>
```

For example:

```
man grep
```

or:

```
grep --help
```

That habit is especially useful when you start working with **Docker, Kubernetes, and AWS servers**.